



Paper ID : 261208

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Subject Code:BP704T

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BPHARM
(SEM. VII) THEORY EXAMINATION 2025-26
NOVEL DRUG DELIVERY SYSTEM (NDDS) – THEOR

TIME: 3 HRS**M.MARKS: 75**

Note: 1. Attempt all Sections. If you require any missing data, then choose suitably.

SECTION A

1. Attempt all questions in brief.

10 x 2 = 20

a.	State the drug delivery approaches of design-controlled release formulations based on dissolution.
b.	Classify polymers used in controlled-release drug delivery formulations.
c.	State the merits and demerits of microspheres.
d.	Mention the limitations of the osmotic pump.
e.	Draw a neat, labeled sketch representing the layers of skin for drug permeation.
f.	State the working principle of nebulizers.
g.	Name to major targeting ligands for liposomes and their targets.
h.	Define monoclonal antibodies and their applications.
i.	State the requirements of an ophthalmic drug for intracorneal permeation.
j.	Name and state the mechanisms of any two major IUDs.

SECTION B

2. Attempt any two parts of the following:

2 x 10 = 20

a.	State and explain the physicochemical and biological properties of drugs relevant to controlled release formulations.
b.	Discuss the various classes of permeation enhancers, mentioning the merits, limitations, and applications of each class.
c.	State and explain the intraocular barriers and the formulation approaches to overcome them.

SECTION C

3. Attempt any five parts of the following:

7 x 5 = 35

a.	Explain the method of selection of drug candidates for controlled release formulations.
b.	Discuss the formulation considerations for buccal drug delivery systems.
c.	Explain in brief the formulation concept of pharmaceutical implants along with their applications.
d.	Describe the various approaches for GRDDS along with their applications.
e.	Classify inhalers and suggest the formulation approaches of each class.
f.	Explain the method of preparation of monoclonal antibodies with a neat schematic diagram.
g.	Write a short note on ocuserts.